

AARON SUN

Experience

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Plan Group

May 2026 – August 2026

Electrical Designer

Vaughan, Ontario

- Developed and modified **electrical building designs in Revit**, working within the power-design portion of project teams to produce coordinated electrical layouts and documentation.
- Created and updated **power distribution drawings and electrical layouts**, incorporating equipment, branch circuits, and distribution components into building designs.
- Coordinated electrical designs within Revit to maintain accurate **equipment locations, circuit layouts, and drawing documentation** across project revisions.
- Reviewed electrical drawings and design changes with the project team, gaining experience working within **engineering constraints, design standards, and multidisciplinary projects**.

Avnan

September 2025 – December 2025

Electronics Engineering Technician

Mississauga, Ontario

- Performed hands-on **PCB rework and repair** on production and prototype boards, including SMD and through-hole soldering, component replacement, and inspection under magnification.
- Conducted **functional and continuity testing** on assembled electronic products to identify manufacturing defects and verify circuit integrity.
- Supported the hardware team in **manufacturing validation and quality assurance**, gaining practical experience with electronics assembly and failure modes.

New Radio Tech Inc.

January 2025 – April 2025

Assistant Engineer

Shenzhen, Guangdong

- Refined automated **Python test scripts**, modularizing functions and adding error handling to reduce average test cycle time by **18%**.
- Performed bench-level electrical testing on **50+ prototype microchips**, using oscilloscopes, power supplies, and custom test fixtures to validate performance and isolate faults.
- Analyzed high-volume semiconductor test data using **Python, Pandas, and NumPy**, generating yield reports and trend visualizations to support root-cause analysis.
- Collaborated with design, firmware, and manufacturing teams to improve testing procedures and document standardized test processes.

Projects

RISC-V CPU | *SystemVerilog, Python, Yosys, Verilator*

September 2026

- Designed and implemented a **single-cycle RV32I CPU** in SystemVerilog, including the ALU, register file, control decoder, immediate generator, and memory subsystem.
- Built a **layered verification suite** (per-module unit tests, a 24-check full-core instruction regression, and a randomized differential test against a Python golden model) achieving a **100% pass rate across 10,000+ randomized ALU instructions**.
- Wrote a **two-pass Python assembler** with label resolution to generate test programs, eliminating hand-computed branch/jump offsets.
- Verified design correctness using **Verilator lint** and synthesized the design with **Yosys**, confirming a clean 121-cell gate-level netlist with no synthesis errors.

RC Car | *C, Arduino, KiCad, AutoCAD*

May 2025 – July 2025

- Co-engineered an Arduino-based **4WD RC car and wireless handheld controller**, integrating two Arduino MCUs, dual L298N motor drivers, four DC motors, battery packs, and nRF24L01 wireless modules.
- Designed the electrical system and **KiCad wiring/harness layout**, integrating motor drivers, power distribution, microcontrollers, and wireless communication hardware.
- Programmed a bidirectional **nRF24L01 wireless communication system in C**, achieving low-latency control over a 30 m line-of-sight range.

Technical Skills

Hardware: PCB Design, Schematic Capture, PCB Layout, SMD & Through-Hole Soldering, STM32, Arduino

EDA / Design: Altium Designer, KiCad, Revit, AutoCAD, ERC/DRC

Programming: C/C++, Python (Pandas, NumPy, Matplotlib), Verilog, SystemVerilog, VHDL

Tools: Git, GitHub, Linux, VS Code

Education

University of Waterloo

September 2024 – Present

Bachelor of Applied Science in Electrical Engineering, President's Scholarship of Distinction

Waterloo, Ontario